



Shield Insurance Business Analytics

Power BI Dashboard & Stakeholder Presentation — Project Report

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Program: Codebasics Data Analyst Virtual Internship — AtliQ Technologies Pvt. Ltd.

Client Stakeholder: Mathew, Business Analyst — Shield Insurance

Project Duration: August 1 - August 14, 2026

Tools: Power BI, DAX, Power Query, PowerPoint

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1. Executive Summary

This report documents the complete lifecycle of the Shield Insurance Business Analytics project — from the original client requirement email to a published, interactive Power BI dashboard and a stakeholder-ready video presentation. Unlike a typical dashboard-only exercise, this project was approached from a Business Analyst perspective: starting with a real (simulated pilot) client brief, translating it into a structured, prioritized feature list, locking mockup designs before development, building a verified data model and DAX layer, and finally converting the analysis into a business narrative delivered through a PowerPoint deck and a recorded video walkthrough.

The analysis covers four dimensions of Shield Insurance's business: overall revenue and customer performance, sales channel (mode) performance, city/geographic concentration, and age-group customer behavior. Using six months of transactional data (November 2022 – April 2023, 26,841 policy-sale records), the dashboard surfaces four headline findings:

- 1. Strong overall growth** — ₹989M revenue from 26,841 customers, both up ~18% month-over-month.
- 2. Channel mix is shifting** — Offline-Agent remains the largest channel (55.67% of revenue) but its monthly share has declined steadily from 63.6% to 44.8%, while digital channels grew from 6.1% to 35.0% over the same period.
- 3. Delhi NCR is the leading market** — ₹402M revenue from 11,007 customers, the highest of all five cities.
- 4. The 55+ age group is a high-value segment** — despite being only 12.8% of customers, it delivers 27.8% of revenue, the highest average premium (₹79,635), and the highest expected settlement rate (71.98%).

The complete deliverable set includes: a 4-page interactive Power BI dashboard (Home, General View, Sales Mode Analysis, Age Group Analysis), a full DAX measure library, a 7-slide stakeholder presentation with a matching speaking script, and a recorded video walkthrough — all traceable back to the original client requirement.

2. Business Problem & Client Brief

The project began as a pilot engagement for Shield Insurance, communicated via a requirement email from **Mathew, Business Analyst at Shield Insurance**. The brief (summarized from the original email) requested the following:

"To begin, we would like to focus on understanding the number of customers we have and the total revenue we are generating. It would also be beneficial to track the daily revenue growth rate and daily customer growth rate to monitor our progress.

Monitoring changes in policies on a month-over-month basis is also important to identify trends and areas for improvement. It would be helpful to segment our customer base by age group and analyze revenue and customer numbers by city and age group.

To analyze trends in customer and revenue growth over time, it would be great to create a switch between revenue trend graphs and customer trend graphs. Additionally, using filters to analyze sales mode, age group, city, month, and policy ID would make the analysis more efficient.

It would be valuable to have a separate page for sales mode analysis... We can calculate total customers and total revenue split percentages by sales mode and analyze the trend of sales mode over the month.

Having a separate page for age group analysis would also be helpful... We can analyze age group data to understand expected settlement, sales mode, and policy preference, which will help us make informed business decisions."

— Mathew, Business Analyst, Shield Insurance

The client also confirmed that this was a **pilot project** intended to build confidence in the working relationship, with the data and metadata attached for reference.

3. Objective

The objective was not simply to produce a dashboard, but to run the analysis through a complete Business Analyst workflow:

Interpret the client's unstructured business email



Convert it into a structured, prioritized feature list



Design and lock page-level mockups before any development



Build a verified data model, DAX layer, and 4-page dashboard



Translate dashboard findings into a business narrative (PPT + script)



Deliver via video walkthrough and published Power BI report

4. Feature List & Requirement Traceability

Before any dashboard element was built, the client email was decomposed into a structured feature list (17 features, each with a business justification and priority level). This step ensured every visual on the final dashboard could be traced back to an explicit business need — rather than being added arbitrarily.

#	Feature	Business Justification	Priority
1	Total Customers, Total Revenue, Daily Revenue Growth, Daily Customer Growth as key metrics	To monitor all crucial metrics at a glance	High
2	Month-over-month % change on key metrics	Tracking MoM change in policies offers insight into performance trends	High
3	Segment customers by age group (18-24, 25-30, 31-40, 41-50, 51-65, 65+)	To categorize customers and understand behavior/preferences	High
4	Total revenue split by age group and city	To identify the most profitable demographics and geographies	High
5	Total customers split by age group and city	To customize products and marketing by location and age	High
6	Customer / daily customer growth trend by month	To track growth trends and spot behavior changes over time	Low
7	Revenue / daily revenue growth trend by month	To track financial performance and spot fluctuations	Low
8	Toggle between revenue trend and customer trend graphs	Improves UX by letting users switch views efficiently	Low
9	Filters on sales mode, age group, city, month, policy ID	Enables targeted, efficient analysis	High
10	Separate page for sales mode analysis	A dedicated page for channel analysis	High
11	Total customers split % by sales mode	Shows which channels attract/retain customers best	High
12	Total revenue split % by sales mode	Shows which channels generate the most revenue	High
13	Trend of sales mode over month	To understand how channel mix evolves monthly	Low
14	Separate page for age group analysis	A dedicated page for demographic analysis	High
15	Age group vs. expected settlement	To understand expected annual settlement by segment	High
16	Age group vs. sales mode preference	To understand channel preference by age group	Low
17	Age group vs. policy preference	To understand which policies are bought by which age group	High

5. Dataset & Metadata

Five CSV tables were provided by the client, covering customer, date, policy, premium (sales), and settlement data.

Table	Rows	Key Columns	Description
dim_customer	26,841	customer_code, dob, city	Customer master data
dim_date	182	date, mmm_yy, day_type, week_no	Calendar table (Nov 2022 – Apr 2023)
dim_policies	9	policy_id, base_coverage_amt, base_premium_amt	Policy master data — no policy-type/category column
fact_premiums	26,841	date, customer_code, policy_id, sales_mode, final_premium_amt	Transactional policy-sale records
fact_settlements	73	age, settlement %	Standalone table — settlement rate by single age (18–90), no date or count field

Date range covered: 01-Nov-2022 to 30-Apr-2023 (6 months, labeled M1-M6 on trend charts).
Sales modes in raw data: Offline-Agent, Offline-Direct, Online-App, Online-Website.
Cities in raw data: Mumbai, Hyderabad, Delhi NCR, Chennai, Indore.

Data Confidentiality Note

Per internship program guidelines, the underlying raw datasets (CSV files) are not shared publicly on GitHub or social platforms. This report and the accompanying GitHub repository include only the data model design, DAX logic, dashboard screenshots, and derived (aggregated) figures — not the source records.

6. Tools & Technology

Tool	Purpose
Power BI Desktop	Data modeling, DAX measures, report/visual build, publishing
Power Query	Data cleaning and transformation (age-group buckets, calculated columns)
DAX	Time-intelligence measures, TREATAS filter propagation, conditional formatting logic
Power BI Service	Publishing the report for stakeholder access
Microsoft PowerPoint	7-slide stakeholder presentation deck
Screen recording / video	Recorded walkthrough presentation for asynchronous stakeholder review

7. Project Workflow

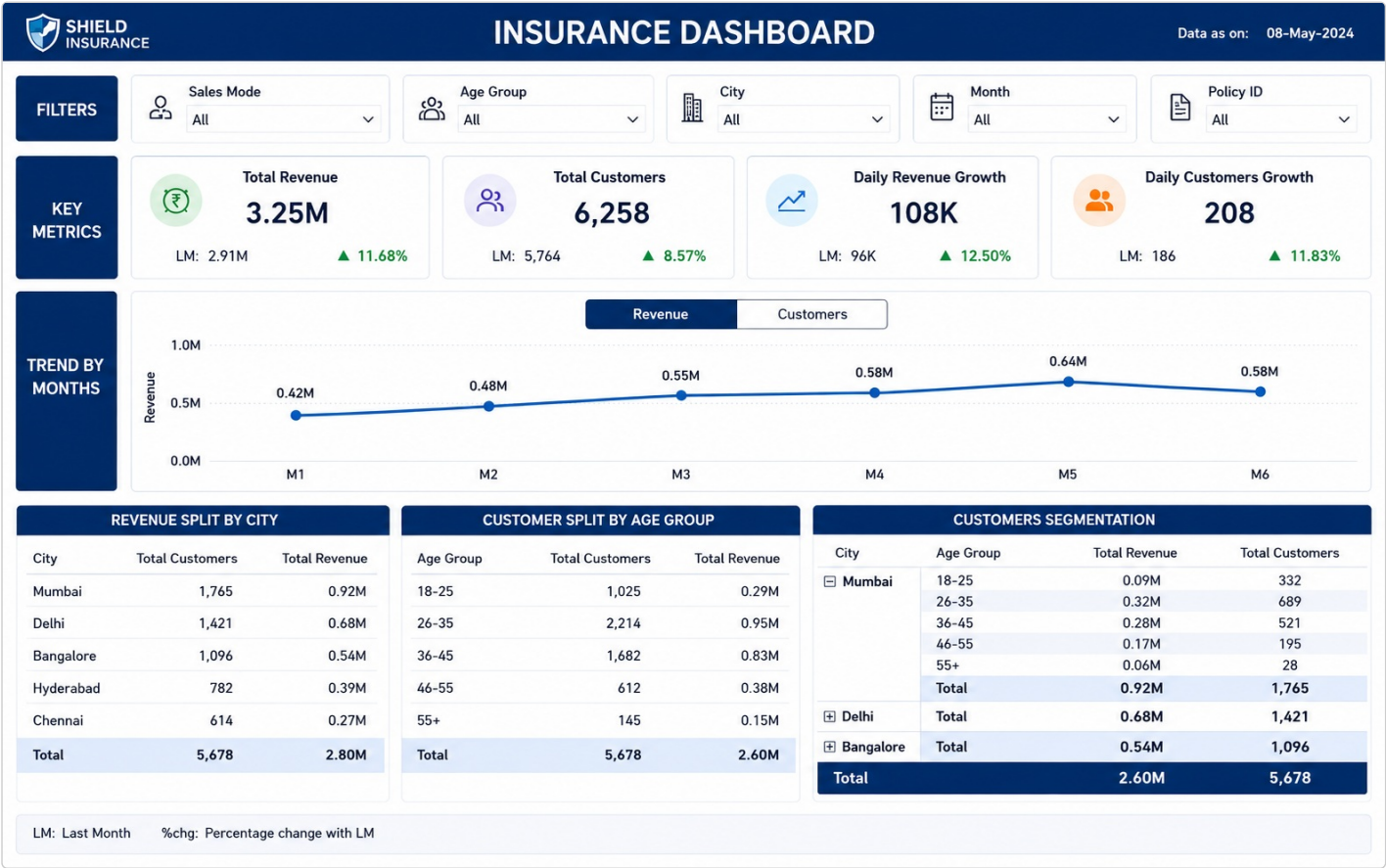
Development followed a strict, incremental, mockup-first methodology rather than building visuals ad hoc:

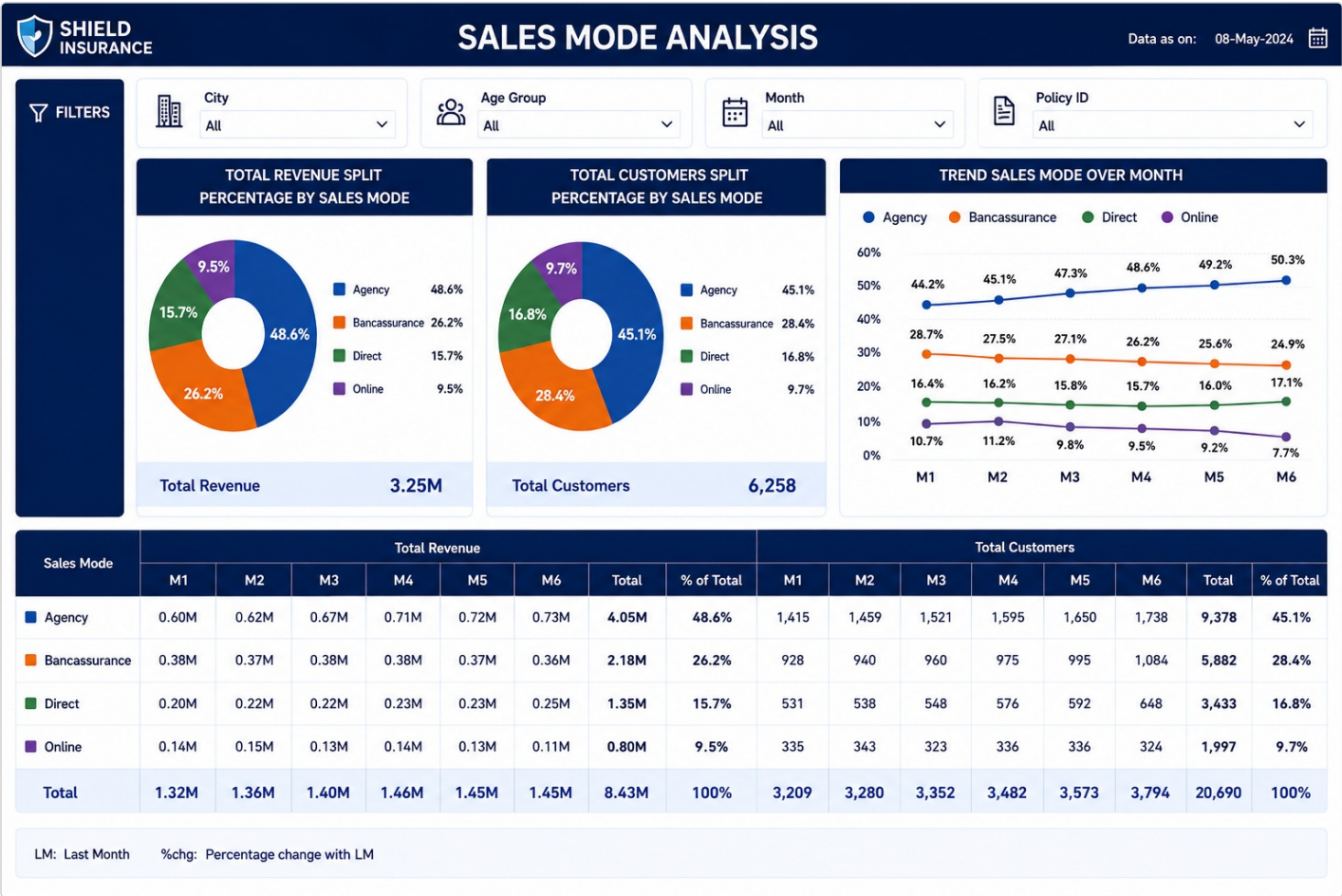
Step	Activity
1	Read and interpret the client email (Mathew, Shield Insurance)
2	Build a structured, prioritized feature list (17 features) directly traceable to the email
3	Design and lock three page-level mockups (General View, Sales Mode Analysis, Age Group Analysis) before any development began
4	Clean and verify all five source tables in Power Query
5	Build the star-schema data model and establish relationships
6	Build the full DAX measure library (Total Revenue, Total Customers, LM/time-intelligence measures, % change with arrows, TREATAS-based settlement measure, ISINSCOPE-based grand-total %)
7	Build each visual incrementally: <i>Select Visual → Identify Measures → Create Measures → Build → Format → Test → Next Visual</i>
8	Apply brand-consistent design system (navy/white theme, 16:9 canvas, 8px-multiple spacing)
9	Apply conditional formatting (green/red, up/down arrows) to all KPI cards based on MoM % change
10	Publish the finished report to Power BI Service
11	Design a 7-slide stakeholder presentation translating dashboard findings into a business narrative
12	Write and align a word-for-word speaking script to the presentation's visual hierarchy
13	Record and publish a video walkthrough

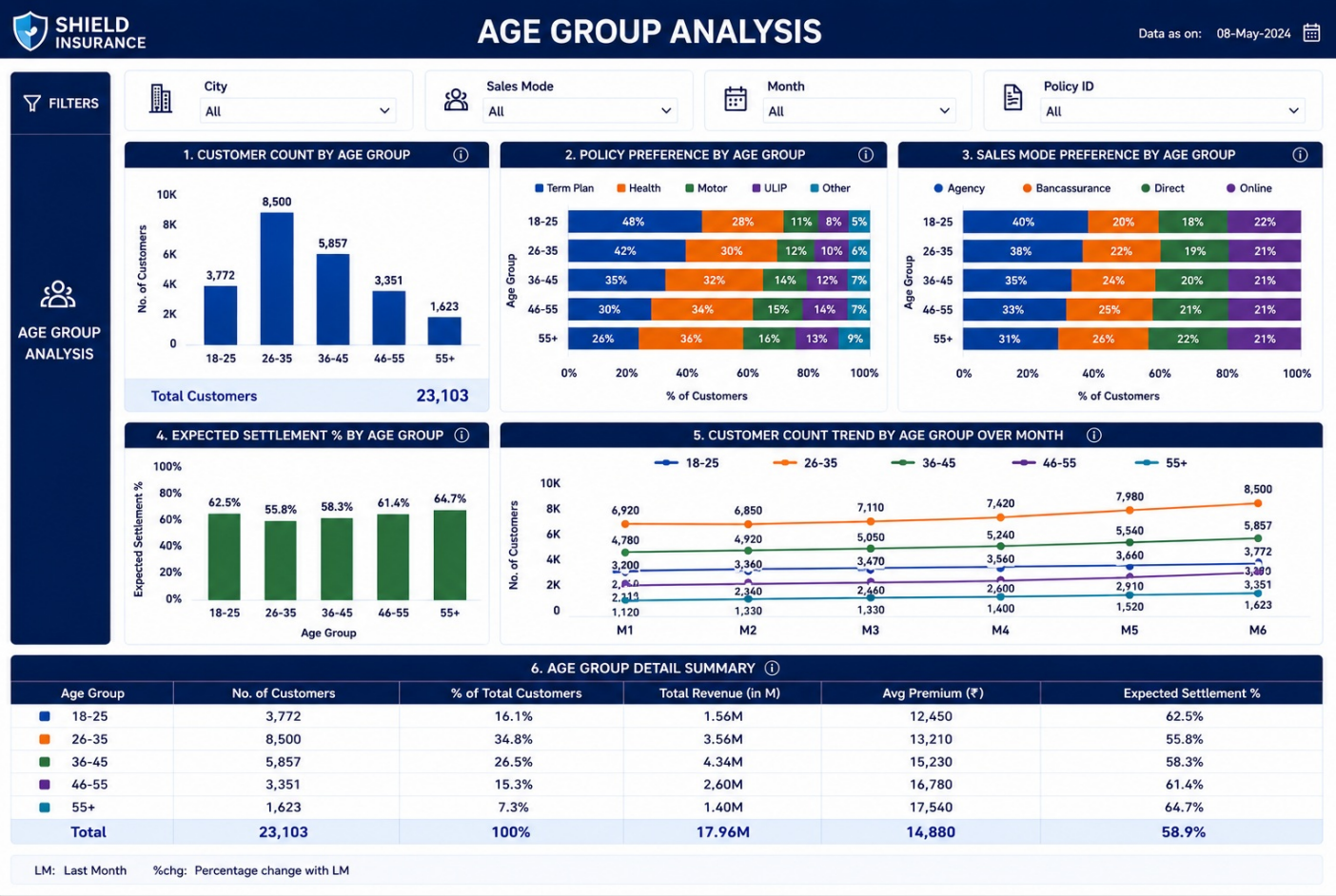
8. Mockup Design (Locked Wireframes)

Before development began, three page-level mockups were designed and explicitly locked as the build target. All layout, KPI card placement, filter positions, and visual numbering shown below were treated as the specification that superseded any later ad hoc changes.

Page 1 — General View (Mockup)







9. Data Model & Relationships

A star-schema model was built with four dimension/fact tables physically related, and the settlement table intentionally left disconnected due to its structure.

Relationship	Type
dim_customer[customer_code] ↔ fact_premiums[customer_code]	One-to-many, active
dim_date[date] ↔ fact_premiums[date]	One-to-many, active (marked as Date Table)
dim_policies[policy_id] ↔ fact_premiums[policy_id]	One-to-many, active
fact_settlements	No physical relationship — disconnected table, filter context applied via DAX TREATAS ()

Key Data Integrity Decision

fact_settlements

contains only **age** and **settlement %** — no date, count, or amount column. This makes settlement trends over time structurally impossible with this table. The "Customer Count Trend by Age Group" visual therefore uses

fact_premiums

+

dim_customer

+

dim_date

instead, while settlement analysis is limited to a static (non-time-based) view, joined to the model via a calculated Age Group column and TREATAS.

10. DAX Measures — Full Reference

All measures below were built and verified in the Measure Table.

Total Revenue

```
Total Revenue = SUM(fact_premiums[final_premium_amt(INR)])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(fact_premiums[customer_code])
```

Total Revenue LM (Last Month)

```
Total Revenue LM =  
CALCULATE(  
    [Total Revenue],  
    DATEADD(dim_date[date], -1, MONTH)  
)
```

Total Customers LM

```
Total Customers LM =  
CALCULATE(  
    [Total Customers],  
    DATEADD(dim_date[date], -1, MONTH)  
)
```

Revenue % Change

```
Revenue % Change =  
DIVIDE(  
    [Total Revenue] - [Total Revenue LM],  
    [Total Revenue LM]  
)
```

Revenue % Change with Arrow

```
Revenue % Change with Arrow =  
VAR PctChange = [Revenue % Change]  
VAR ArrowIcon = IF(PctChange >= 0, "▲ ", "▼ ")  
RETURN  
    ArrowIcon & FORMAT(PctChange, "0.00%")
```

Customers % Change

```
Customers % Change =  
DIVIDE(  
    [Total Customers] - [Total Customers LM],  
    [Total Customers LM]  
)
```

Customers % Change with Arrow

```
Customers % Change with Arrow =  
VAR PctChange = [Customers % Change]  
VAR ArrowIcon = IF(PctChange >= 0, "▲ ", "▼ ")  
RETURN ArrowIcon & FORMAT(PctChange, "0.00%")
```

Daily Revenue Growth

```
Daily Revenue Growth =  
DIVIDE(  
    [Total Revenue],  
    DISTINCTCOUNT(dim_date[date])  
)
```

Daily Revenue Growth LM

```
Daily Revenue Growth LM =  
CALCULATE(  
    [Daily Revenue Growth],  
    DATEADD(dim_date[date], -1, MONTH)  
)
```

Daily Revenue Growth % Change (with Arrow)

```
Daily Revenue Growth % Change =
```

```

DIVIDE(
    [Daily Revenue Growth] - [Daily Revenue Growth LM],
    [Daily Revenue Growth LM]
)

Daily Revenue Growth % Change with Arrow =
VAR PctChange = [Daily Revenue Growth % Change]
VAR ArrowIcon = IF(PctChange >= 0, "▲", "▼")
RETURN ArrowIcon & FORMAT(PctChange, "0.00%")

```

Daily Customers Growth (+ LM, % Change, Arrow)

```

Daily Customers Growth =
DIVIDE(
    [Total Customers],
    DISTINCTCOUNT(dim_date[date])
)

Daily Customers Growth LM =
CALCULATE(
    [Daily Customers Growth],
    DATEADD(dim_date[date], -1, MONTH)
)

Daily Customers Growth % Change =
DIVIDE(
    [Daily Customers Growth] - [Daily Customers Growth LM],
    [Daily Customers Growth LM]
)

Daily Customers Growth % Change with Arrow =
VAR PctChange = [Daily Customers Growth % Change]
VAR ArrowIcon = IF(PctChange >= 0, "▲", "▼")
RETURN ArrowIcon & FORMAT(PctChange, "0.00%")

```

Avg Premium

```

Avg Premium = AVERAGE(fact_premiums[final_premium_amt(INR)])

```

% of Total Customers (by Age Group)

```

% of Total Customers =
DIVIDE(
    [Total Customers],
    CALCULATE([Total Customers], ALL(dim_customer[Age Group]))
)

```

Avg Settlement % (simple)

```

Avg Settlement % = AVERAGE(fact_settlements[settlement %])

```

Avg Settlement % by Age Group — TREATAS Solution

Solves filter propagation into the disconnected

```

fact_settlements

```

table by mapping the Age Group context from

```

dim_customer

```

onto the settlement table's own Age Group column.

```

Avg Settlement % by Age Group =
CALCULATE(
    AVERAGE(fact_settlements[settlement %]),
    TREATAS(
        VALUES(dim_customer[Age Group]),
        fact_settlements[Age Group]
    )
)

```

Revenue % of Total by Sales Mode

```

Revenue % of Total by Sales Mode =
DIVIDE(
    [Total Revenue],
    CALCULATE([Total Revenue], ALLSELECTED(fact_premiums[sales_mode]))
)

```

Customers % of Total by Sales Mode

```
Customers % of Total by Sales Mode =
DIVIDE(
    [Total Customers],
    CALCULATE([Total Customers], ALLSELECTED(fact_premiums[sales_mode]))
)
```

Revenue % of Grand Total — ISINSCOPE Trick

Shows the % only on the "Total" row of the Sales Mode detail table, and blank on individual month columns.

```
Revenue % of Grand Total =
IF(
    ISINSCOPE(dim_date[mmm_yy]),
    BLANK(),
    DIVIDE(
        CALCULATE([Total Revenue], ALL(dim_date[mmm_yy])),
        CALCULATE([Total Revenue], ALLSELECTED(fact_premiums[sales_mode]), ALL(dim_date[mmm_yy]))
    )
)
```

Customers % of Grand Total

```
Customers % of Grand Total =
IF(
    ISINSCOPE(dim_date[mmm_yy]),
    BLANK(),
    DIVIDE(
        CALCULATE([Total Customers], ALL(dim_date[mmm_yy])),
        CALCULATE([Total Customers], ALLSELECTED(fact_premiums[sales_mode]), ALL(dim_date[mmm_yy]))
    )
)
```

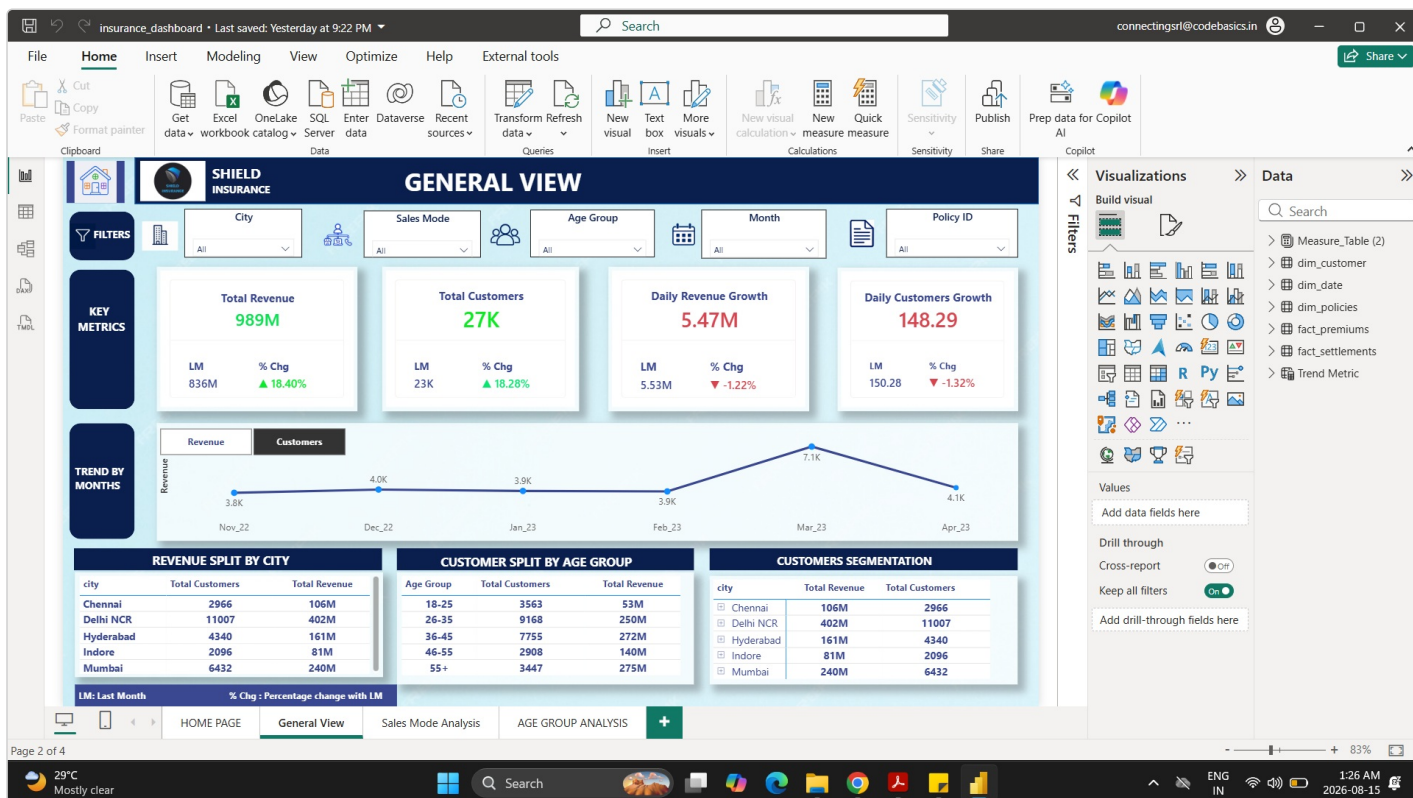
Conditional Formatting Logic (KPI Cards)

Applied via Power BI's rules-based conditional formatting (not hardcoded in DAX), based on the corresponding numeric % Change measure for each KPI card, on both the value and the "with Arrow" label:

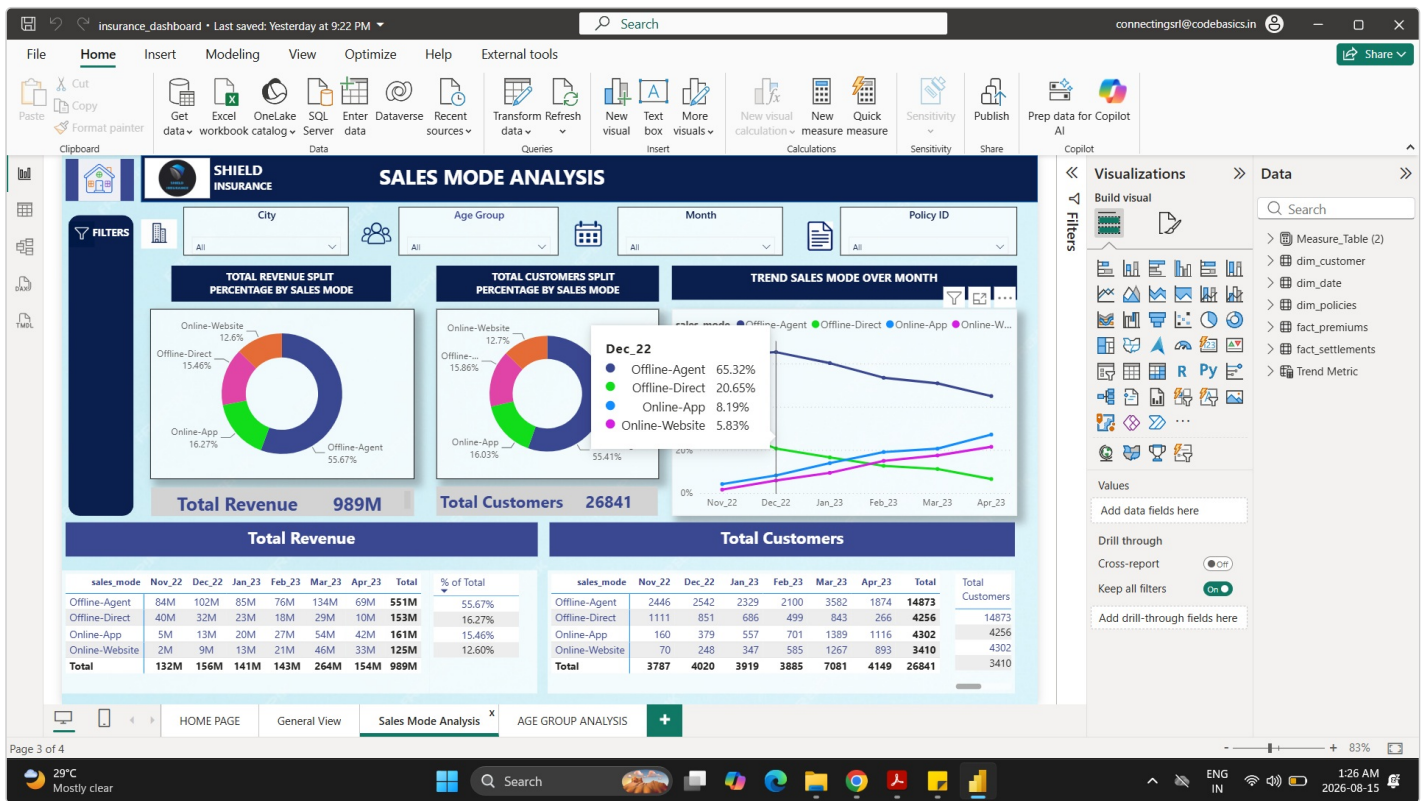
```
Rule 1: IF value >= -1 AND value < 0 THEN color = Red (#D13438)
Rule 2: IF value >= 0 AND value <= 1 THEN color = Green (#107C10)
```

11. Dashboard Pages — Final Build

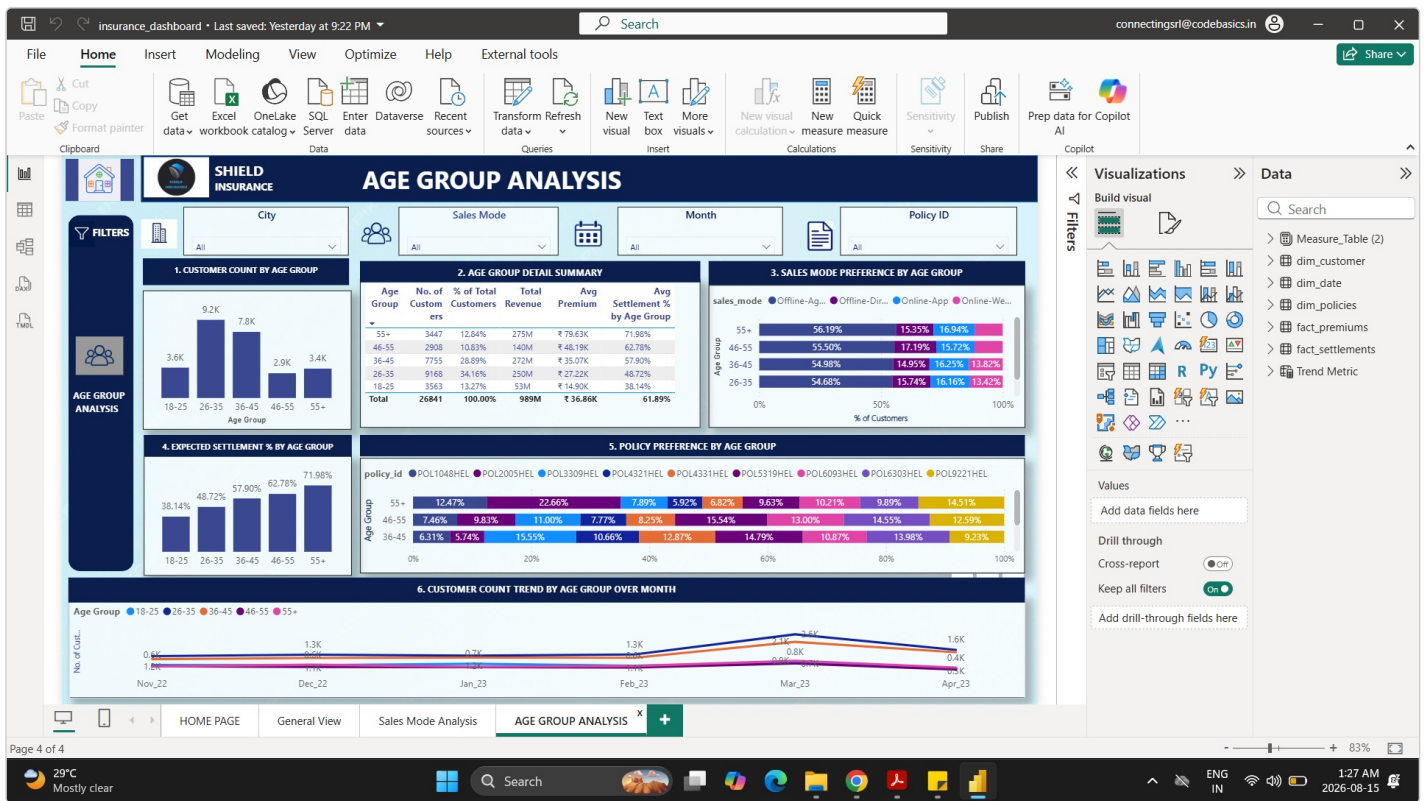
Page 1 — General View



4 conditionally-formatted KPI cards (Total Revenue ₹989M, Total Customers 27K, Daily Revenue Growth ₹5.47M, Daily Customers Growth 148.29), Revenue/Customer trend toggle, and 3 breakdown tables by city, age group, and city×age-group drill-down.



Revenue and customer split by sales mode (donut charts), monthly trend line chart with interactive tooltip, and detailed month-by-month revenue/customer tables with % of Total.



6-panel layout: Customer Count by Age Group, Age Group Detail Summary table (with Avg Premium and Settlement %), Sales Mode Preference, Expected Settlement %, Policy Preference (by policy_id), and Customer Count Trend Over Month.

12. Business Questions Answered

Business Question	Answer Location
How many customers and how much revenue are we generating?	Page 1 — Key Metrics KPI cards
What does our customer base look like?	Page 1 — Customer Split by Age Group; Page 3 — full profile
Which cities and age groups drive the business?	Page 1 — Revenue Split by City, Customer Segmentation drill-down
Which sales modes drive revenue and customers?	Page 2 — Sales Mode Analysis (donuts, trend, detail table)
How do settlement, policy, and sales-mode preferences vary by age group?	Page 3 — Age Group Analysis (all 6 panels)

13. Key Insights (Quantified)

Insight	Supporting Numbers
Overall growth	Revenue ₹989M (▲18.40% MoM); Customers 26,841 (▲18.28% MoM)
Daily performance moderation	Daily Revenue ₹5.47M (▼1.22%); Daily Customers 148.29 (▼1.32%)
Monthly revenue peak	March 2023 peaked at ₹264M, vs. ₹154M in April
Channel mix shift	Offline-Agent share: 63.6% (Nov) → 44.8% (Apr); Digital (App+Website): 6.1% → 35.0%
Leading market	Delhi NCR: ₹402M revenue, 11,007 customers — highest of 5 cities
Largest customer segment	Age 26-35: 9,168 customers (34.2% of base), ₹250M revenue
Highest-value segment	Age 55+: only 12.8% of customers but 27.8% of revenue; Avg Premium ₹79,635; Expected Settlement 71.98%

14. Business Recommendations

1. Accelerate Digital Growth (Channel)

Use the strong upward movement in Online-App and Online-Website share to expand digital adoption, while maintaining Offline-Agent as an important channel (it remains the largest single channel by revenue).

2. Protect & Grow the 55+ Segment (Customer Value)

The 55+ segment combines the highest average premium (₹79,635) with the highest expected settlement rate (71.98%). Prioritize retention and value-focused offerings for this segment.

3. Deepen Core Markets & Segments (Market)

Continue strengthening Delhi NCR as the leading market, while using the 26–35 segment — the largest customer group by volume — to drive continued customer acquisition and future growth.

Data Note (transparently disclosed to stakeholders): Policy preference is represented by

policy_id

codes rather than named categories, since the source data does not include a policy-type/category column. Settlement data is available only at the age level and does not include a time dimension, so settlement trends over time could not be produced.

15. KPI Impact

KPI	Baseline / Current State	Business Relevance
Total Revenue	₹989M, ▲18.40% MoM	Primary top-line growth indicator
Total Customers	26,841, ▲18.28% MoM	Customer-base expansion indicator
Digital Channel Share	35.0% (up from 6.1% six months prior)	Tracks progress of channel-mix shift toward lower-cost digital acquisition
Revenue Concentration (Delhi NCR)	40.6% of total revenue (₹402M / ₹989M)	Flags geographic concentration risk and expansion opportunity
55+ Segment Value Index	27.8% revenue from 12.8% customers (~2.2x over-index)	Identifies a disproportionately valuable retention target

16. Business Impact

This dashboard equips Shield Insurance stakeholders to move from raw transactional data to specific, defensible decisions: where to invest in digital-channel enablement, which customer segment merits a dedicated retention strategy, and which market to prioritize for continued growth — all grounded in verified, filterable data rather than anecdote. The presentation and video walkthrough were designed so these decisions can be communicated to leadership in under 7 minutes, with every recommendation traceable to a specific chart and number in the underlying dashboard.

17. Stakeholder Presentation (PPT Deck)

A 7-slide, ~7-minute stakeholder presentation was built following an "answer-first, one-message-per-slide" narrative discipline (informed by the "Death by PowerPoint" design philosophy), with the Power BI dashboard serving strictly as supporting evidence rather than being reproduced slide-by-slide.

#	Slide	Core Message
1	Executive Summary	Strong overall growth, with business concentrated in Delhi NCR and the 26–35 segment while sales mix shifts toward digital.
2	Business Questions	The analysis was designed to answer 5 key questions driving revenue, customer, market, and channel decisions.
3	Overall Business Performance	Revenue and customer base grew strongly, but daily performance softened slightly in the latest month.
4	Customer & Market Analysis	Delhi NCR is the largest market, while the 55+ segment contributes disproportionately to revenue.
5	Sales Mode Performance	Offline-Agent remains the largest sales mode, but its share is declining steadily as digital channels gain ground.
6	Customer Behavior by Age Group	Customer value increases with age — 26–35 drives volume, 55+ drives value.
7	Recommendations & Next Steps	Prioritize digital growth, protect high-value customers, and deepen the strongest markets and segments.

The full presentation file (

Shield_Insurance_Executive_Dashboard.pptx

) and a matching, slide-aligned speaking script are included in the project repository.

18. Video Walkthrough & Live Dashboard Links

Deliverable	Link
Video Presentation (YouTube)	https://youtu.be/7iQ8janiMzI
Power BI Report (Service)	Live report link — hosted on personal workspace

Access note: The Power BI Service link points to a personal workspace and requires sign-in / explicit sharing to view. For unrestricted review, the recorded video walkthrough is the primary, publicly accessible demonstration of the dashboard.

19. Known Gaps & Data Limitations

Gap	Detail & Handling
Policy Preference categories	dim_policies has no policy-type/category column, so the Policy Preference visual (Page 3) uses raw policy_id values (9 codes) as the legend instead of named categories such as Term Plan / Health / Motor / ULIP / Other. Accepted as final per project scope; flagged transparently to stakeholders.
Settlement trend over time	fact_settlements contains only age and settlement % — no date or count field — so a time-based settlement trend is not structurally possible with this dataset.
Publication status	Dashboard published to Power BI Service on a personal workspace; broader distribution would require workspace-level sharing or "Publish to Web" (not enabled by default, and not used here to avoid exposing client-referenced data publicly).

20. Future Work

- Add a policy-type/category mapping (in collaboration with the client) to replace raw policy_id codes in the Policy Preference visual.
- Request an updated settlement dataset with a date dimension to enable time-based settlement trend analysis.
- Extend the model with row-level security (RLS) if the report is distributed to multiple regional stakeholders.
- Automate monthly data refresh once Shield Insurance moves from a static extract to a live data source.

21. Author & Contact

Author	Sahajahanur Rahman Laskar
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This report was prepared as part of a Business Analyst / Data Analyst virtual internship pilot project. Raw client datasets are intentionally excluded from this document and from the public repository in line with internship data-confidentiality guidelines.